# The Modernisation Blueprint

## Migration to modern enterprise virtualisation

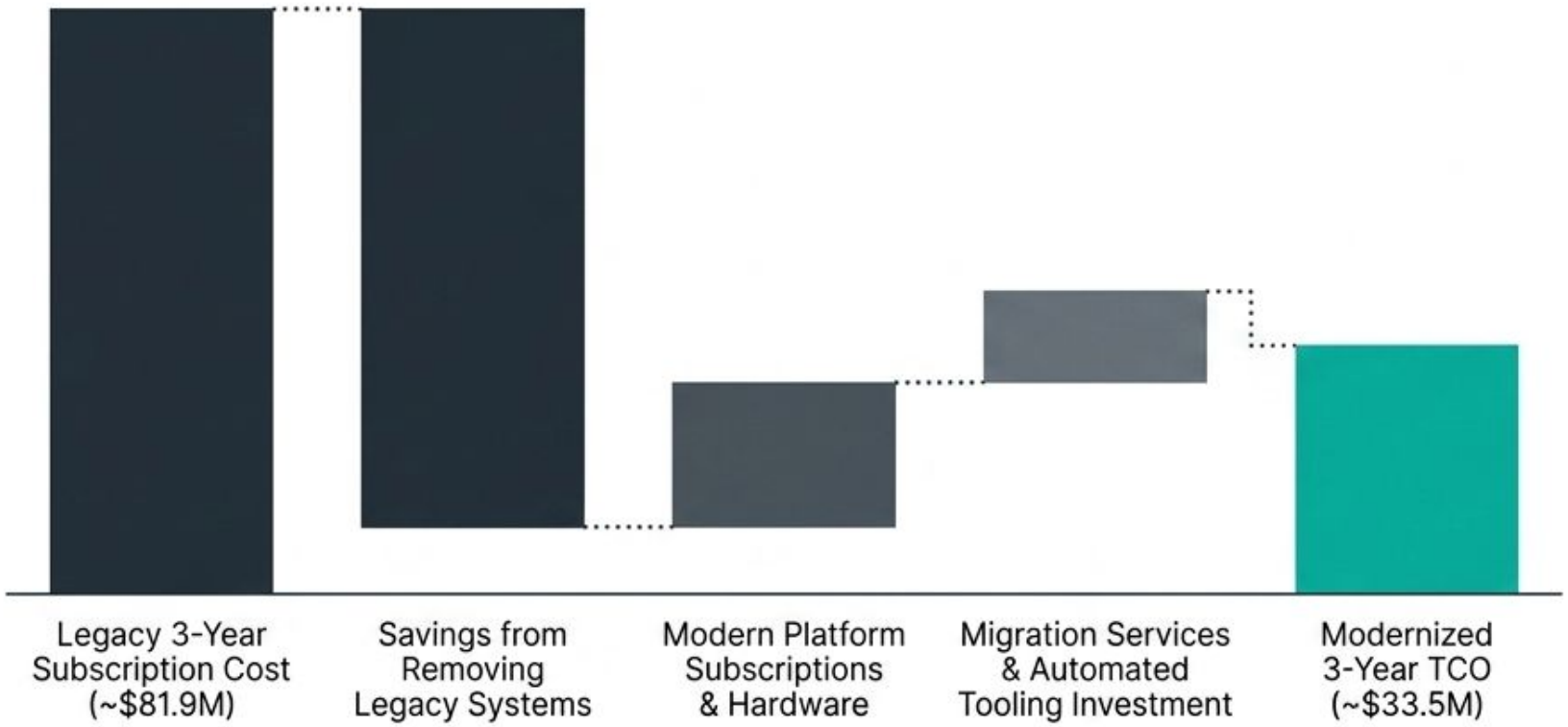


Red Hat

# Redefining the Infrastructure Destination

| | Legacy Virtualization (The Current State) | Container-Native Virtualization (The Future State) |
|---|---|---|
| Architecture | **Siloed Virtual Machines** running on proprietary hypervisors. | **Unified platform** where VMs and Containers coexist seamlessly. |
| Economics | **Escalating, unpredictable** subscription renewals and lock-in. | **Predictable TCO,** leveraging open-source innovation. |
| Operations | **Manual remediation,** disruptive maintenance, and discrete tooling. | **Self-healing automation,** continuous delivery, and unified pipelines. |
| Agility | **Weeks to provision,** scale, and update. | **Declarative provisioning,** instant scaling, and seamless rolling updates. |

# The Financial Reality: Modernization Pays for Itself




**59%**
**Total Cost Reduction**

**$48.3M**
**Net Savings**

over 3 years

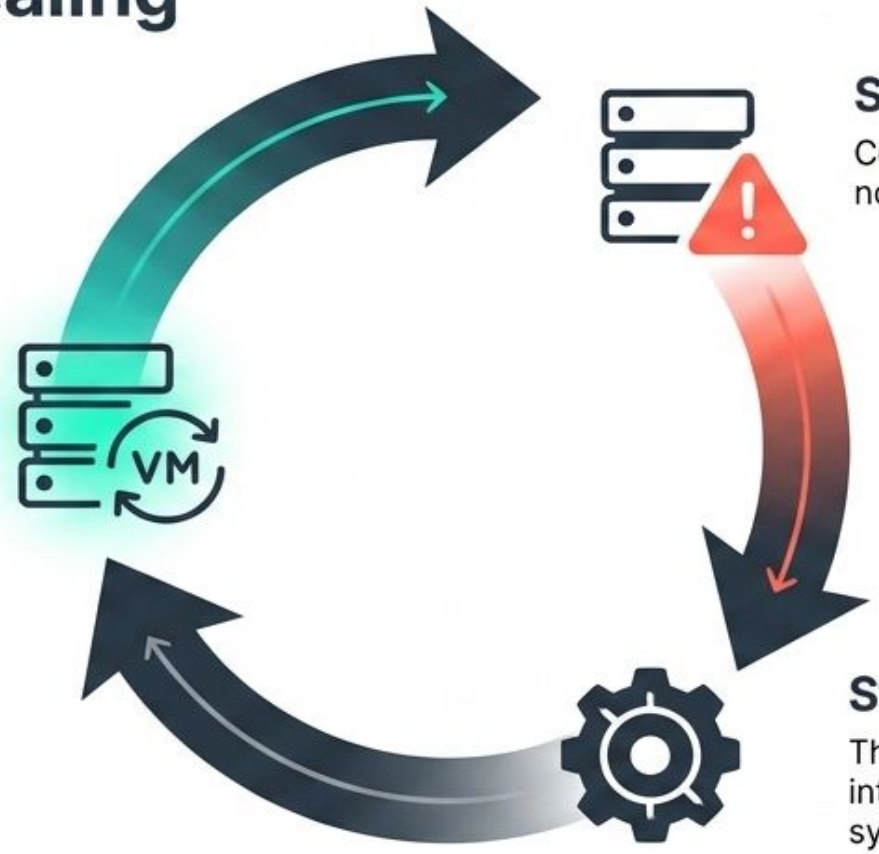
Avoiding escalating legacy renewals fully funds the **migration services**, the new **platform hardware**, and the **software subscriptions**—yielding a massive net positive.

# The Day-2 Advantage: Eradicating Operational Friction

| Mean Time To Recovery (MTTR) | MTTR Rollback |
|---|---|
| 60 minutes → 1 minute | 60 minutes → 1 minute |
| **Time to Market** | **Scalability Provisioning** |
| 20 minutes → 1 minute | 1 week → 1 minute |

In addition to TCO reduction, migrating to a unified application platform protects revenue by drastically reducing unplanned downtime and boosts cost efficiency through accelerated IT productivity.

# The Engine of Resilience: Automated Self-Healing




### Step 1: The Anomaly

Continuous monitoring detects a node or hardware failure instantly.

### Step 2: The Intervention

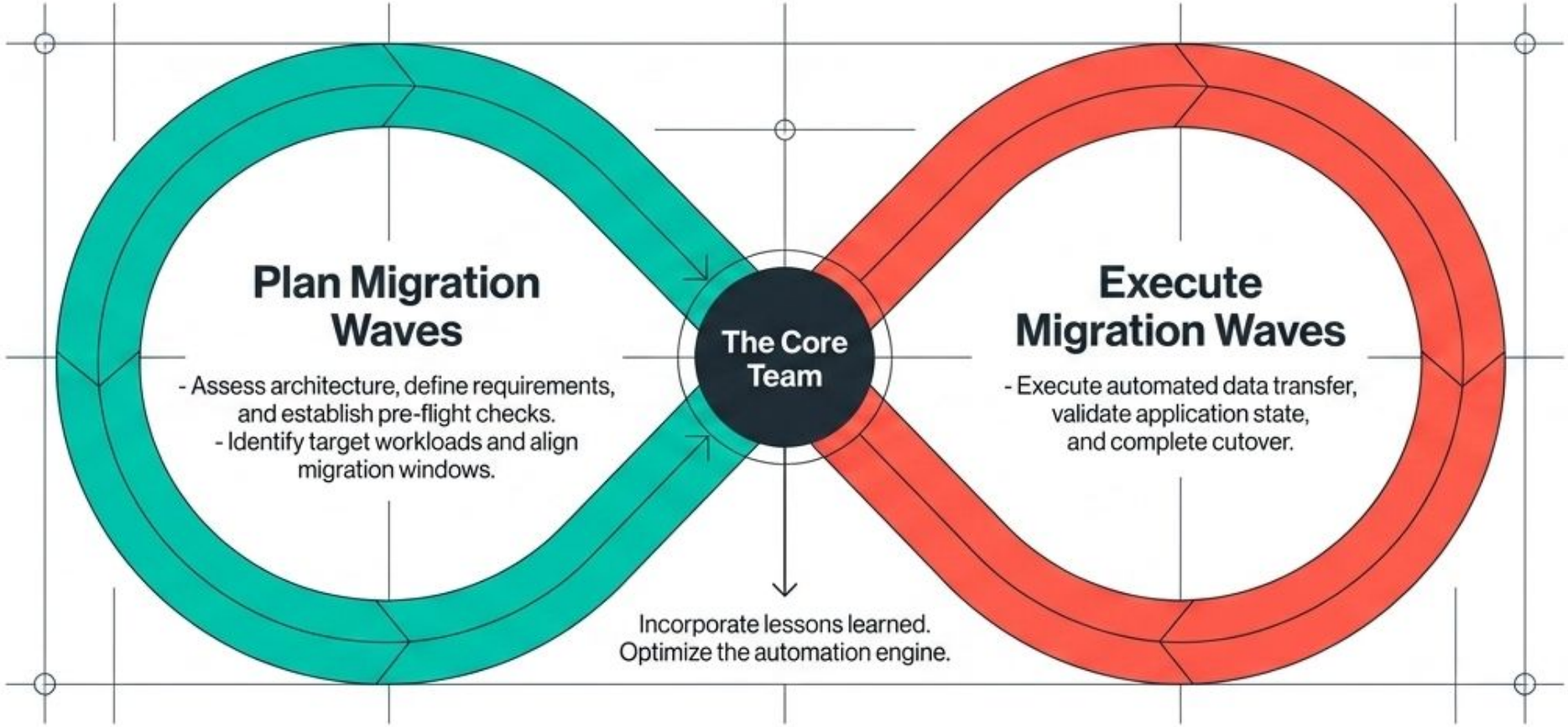
The orchestration engine automatically intercepts the failure—acting as a 24/7 system engineer.

### Step 3: The Resolution

The workload is seamlessly rescheduled and restarted on a healthy node without human intervention.

Traditional environments wait for manual remediation. Modern virtualization continuously enforces application health, ensuring resilience against hardware or network failures.

# The Migration Factory: De-Risking the Journey




**A migration is not a monolithic event. It is a continuous, automated assembly line managed by dedicated Migration Squads, ensuring predictable, non-disruptive execution at scale.**

# Anatomy of an Automated Move

## Phase 0: Discover & Analyze

Automatically detect and catalog workloads, gathering compute, network, and storage prerequisites.

## Phase 1: Automate & Orchestrate

Generate structured inventories, create advanced batch migration plans, and automate the creation of storage/network mapping primitives.

## Phase 2: Steadystate

Post-migration validation, driver installation, and seamless handover to Day-2 ongoing management.

# Mapping the Dependencies: Engineering Predictability

*A migration without a rehearsed, automated rollback capability is just an outage with a project plan.*

## DNS & Certificates

Mitigating total application failure by dual-homing and testing every external dependency (APIs, load balancers) prior to cutover.

## Secrets Management

Eradicating hard-coded credentials by centrally managing and testing all service accounts and database keys.

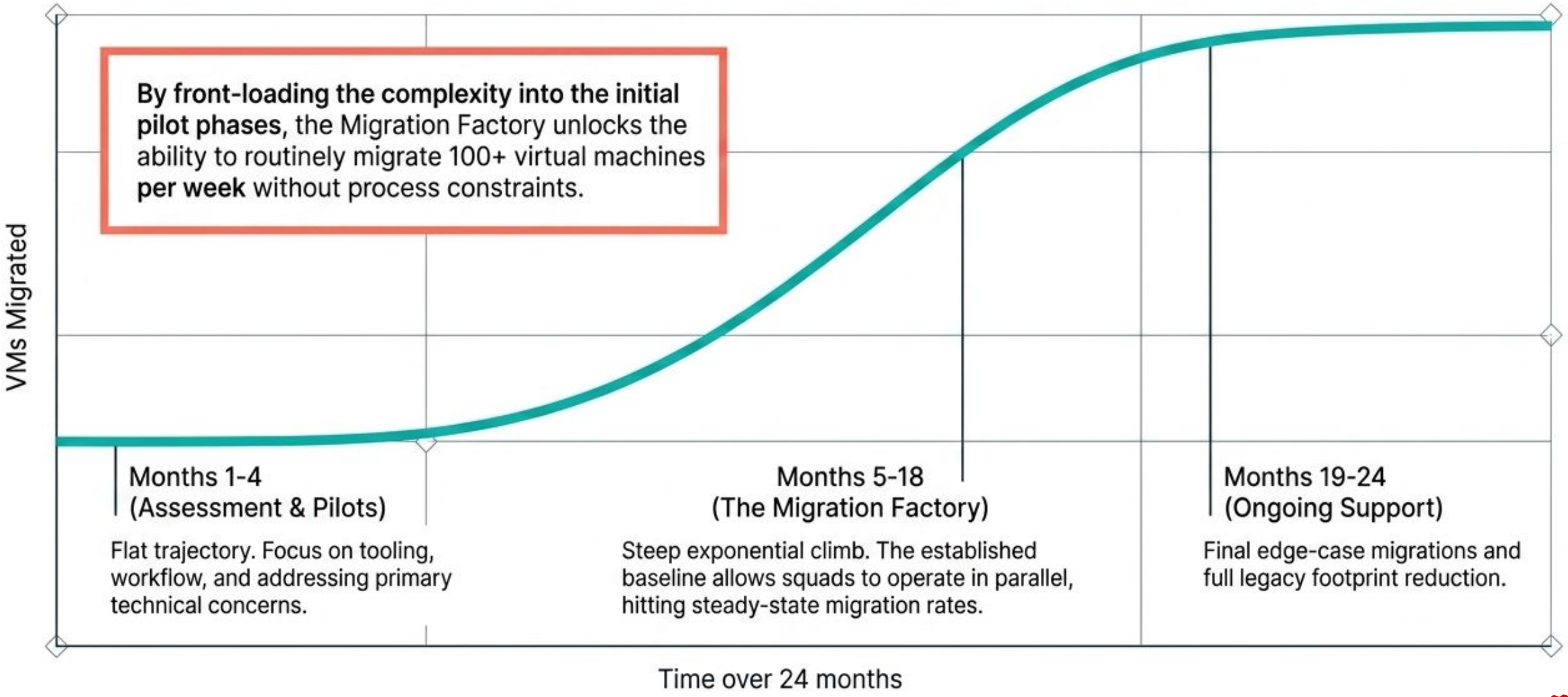
## Configuration Drift

Establishing a reconciled golden source of truth (GitOps) so the new environment remains a trusted, predictable system.

## Monitoring & Alerting

Ensuring every critical health check is configured and tested before a single real user hits the system.

# The Velocity Curve: Exponential Migration at Scale



# The Modernization Synthesis

A Unified Platform that Pays for Itself.

[+]

An Automated Factory that Gets You There Safely.

[=]

**Enterprise Agility at Scale.**

Modernization is not a compromise between cost, safety, and speed. With the right architecture and execution model, they act as force multipliers for each other.

# Proof at Scale:
## The Global Financial Services Migration

### The Organization

A leading global payments technology company processing 13-16 trillion transactions annually.

### The Challenge

A sprawling legacy virtualization footprint across six countries, facing prohibitive licensing costs and a mandate to modernize.

### The Scope

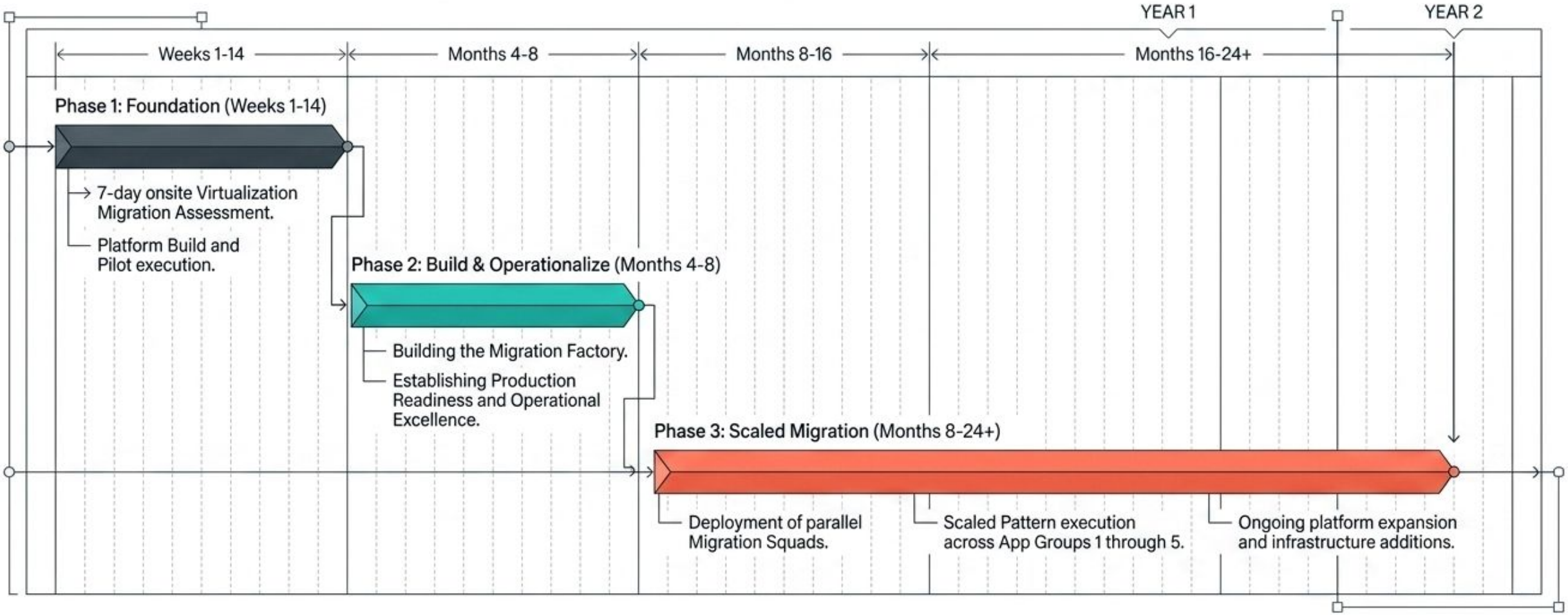
**71,000 Total VMs**
in the global infrastructure.

**66,000 Target VMs**
scoped for migration to the modern container-native platform.

**26,000**
Production VMs.

**33,000**
Non-production VMs.

# Executing the Factory Model: A 24-Month Roadmap



# Validated Outcomes: The Impact of Modernization

*Note: Data validated with a Sample File Processor Application at the target organization.*

| Operational Velocity | Resource Efficiency | Resilience & Agility |
|---|---|---|
| **App Migration Time:** 2 Dev Teams -> 7 Apps migrated seamlessly. | **CPU Consumption:** Reduced from 4X to **1X** | **Mean Time To Recovery (MTTR):** Reduced from 60 minutes to **1 minute** |
| | **Memory Consumption:** Reduced from 50X to **1X** | **Time to Market:** Accelerated from 20 minutes to **1 minute** |

**The greatest operational risk is no longer the migration; it is remaining anchored to legacy architecture.**

# Thank you

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